Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with production infrastructure experience building fleet automation, deployment platforms, and self-healing systems in Python and Go across Kubernetes, Terraform, Ansible, and AWS. Current research builds autonomous LLM agents for automated diagnosis and remediation.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Python, Go, Java, JavaScript, TypeScript, C/C++, SQL.
Infrastructure & Automation: Kubernetes, ArgoCD, Helm, Terraform, Ansible, Docker, Linux, Bash.
Cloud & Platforms: AWS (EC2, S3, Lambda, SQS), HashiCorp Vault, Azure, Google Cloud Platform.
Delivery & Reliability: Git, GitLab CI/CD, Jenkins, CloudWatch, Datadog, SonarQube, REST APIs.
AI & Agents: Gemini, Vertex AI, PyTorch, LangChain, autonomous agent pipelines, root-cause diagnosis.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Built autonomous LLM agent pipeline that diagnoses Android GUI failures from screen recordings and drives automated device remediation, decoupling failure understanding from execution to cut redundant inference.
- Benchmarked against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps, demonstrating **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving diagnosis fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built fleet automation system provisioning, deploying, and retiring **40+ production environments**, migrating process-based deployments to **Kubernetes** via ArgoCD, Helm, GitLab CI/CD, and HashiCorp Vault.
- Designed **Ansible-based Blue-Green deployment** framework with automated rollback, cutting production upgrade downtime by **50%** and enabling self-service environment recovery.
- Automated infrastructure workflows with **Python** (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments; executed **44 production upgrades** in 4 months, root-causing live failures across Linux, EC2, databases, and services using CloudWatch and Datadog telemetry.
- Served as Quality Coach and SonarQube Specialist, raising automated test coverage from **35% to 78%** via CI/CD quality gates; mentored **5 engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** validation framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence-grounded **RAG pipeline** over Google Gemini 2.5-flash, forcing citation-backed answers to eliminate hallucination in high-stakes healthcare eligibility decisions.
- Designed **17-tool MCP server** exposing structured LLM tool-calling for PA extraction, patient-readiness evaluation, and cross-payer comparison, consumable by external AI agents.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).